

Mapping the 3D Parton Structures of Hadrons with Lattice QCD

Dennis Bollweg^{1,2}, Xiang Gao¹, **Jinchen He**^{3,4}, Swagato Mukherjee¹, Peter Petreczky¹, Qi Shi^{1,5} and Yong Zhao⁴

1 Brookhaven National Laboratory, Upton, NY 11973, USA

2 Scientific Computing Core, Flatiron Institute, New York, NY 10010, USA

3 Fermi National Accelerator Laboratory, Batavia IL 60510, USA

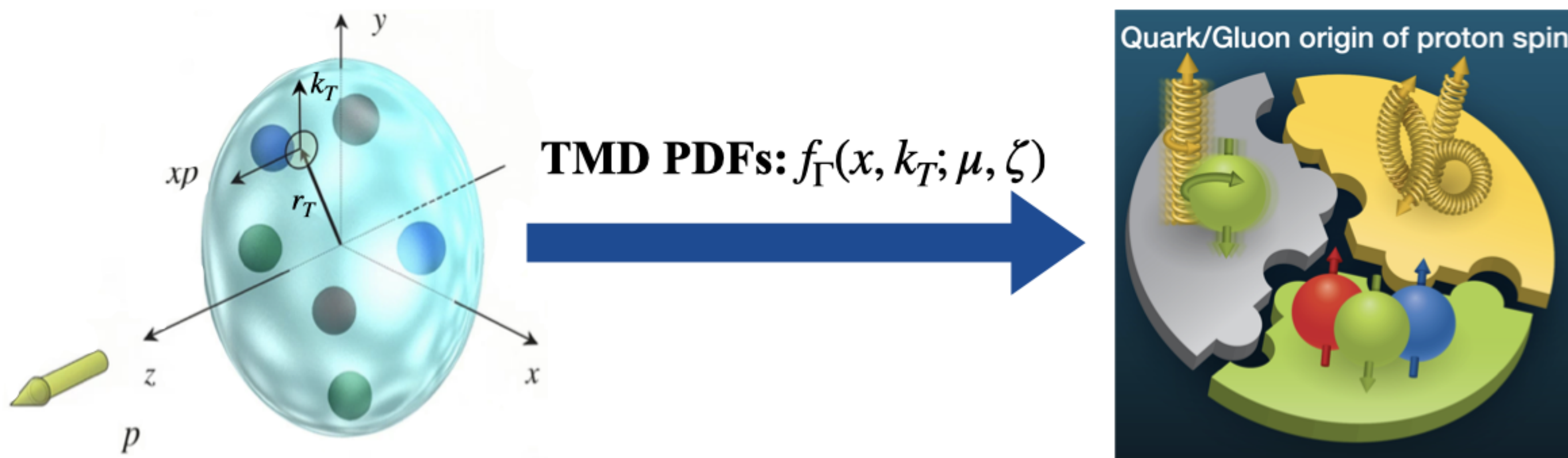
4 Argonne National Laboratory, Lemont, IL 60439, USA

5 Kent State University, Kent, OH 44242, USA

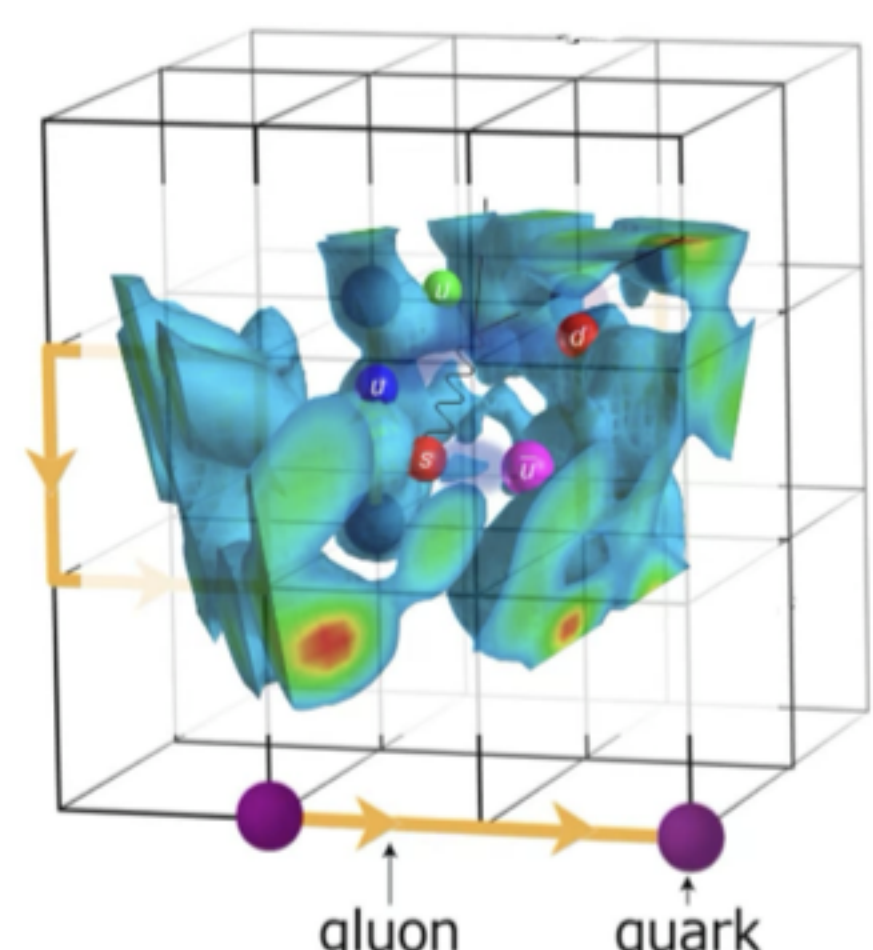
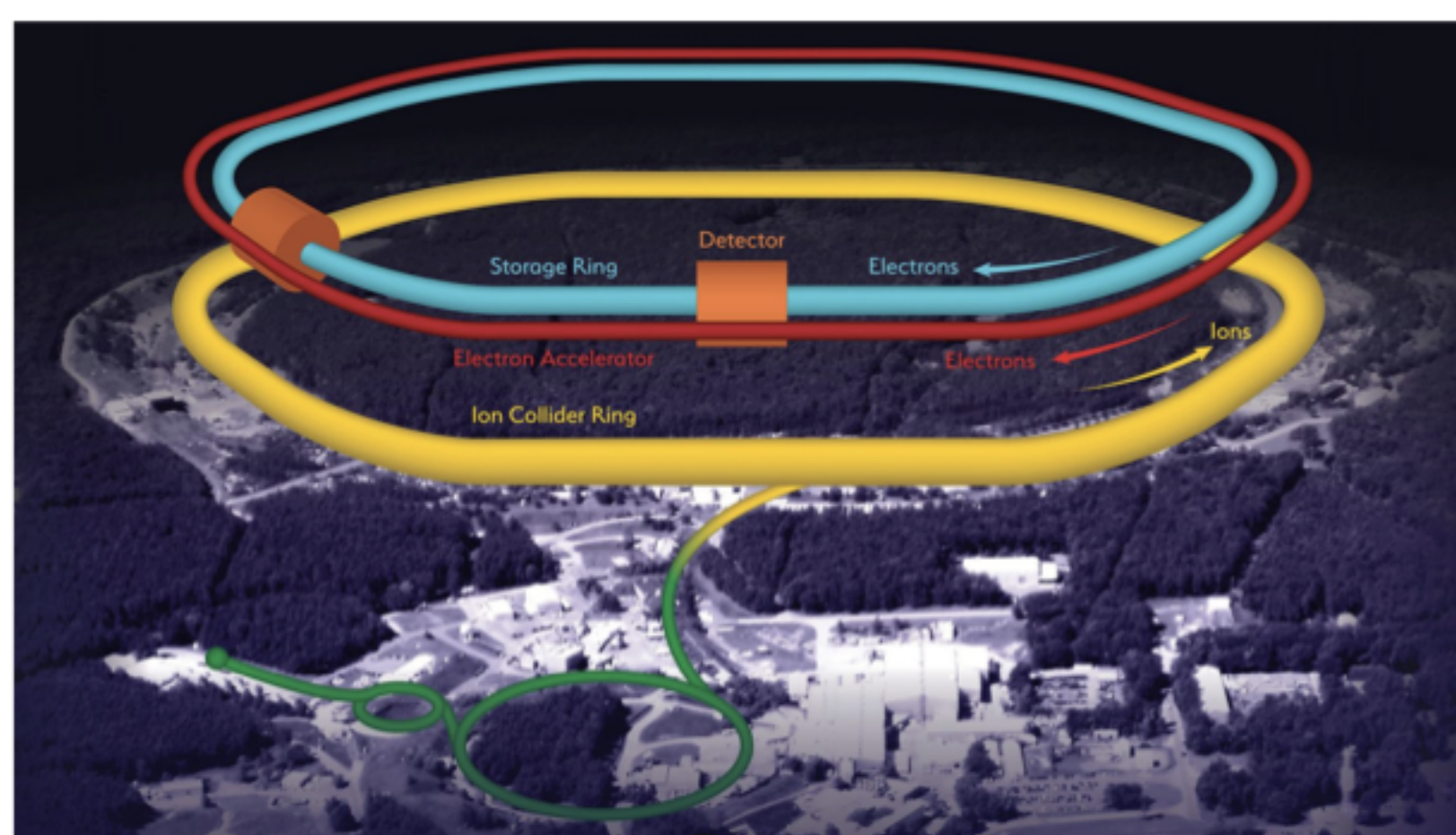
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3D Parton Structures in EIC Era

Parton distribution functions (PDF) and transverse momentum distributions (TMDs) encode the **multidimensional partonic structure** of hadrons, including the **momentum**, spin, and polarization correlations of quarks and gluons.



Lattice QCD provides **first-principles** predictions of hadron structure that complement and guide EIC measurements to reveal the **multidimensional quark and gluon dynamics** inside hadrons.

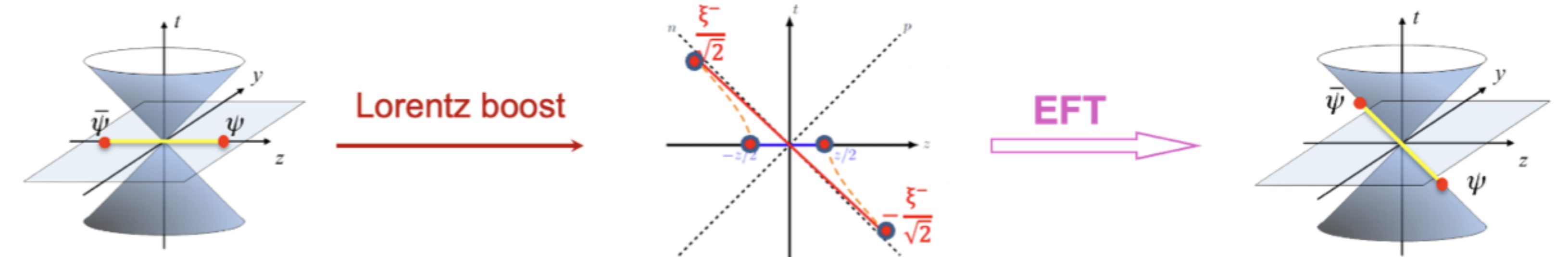


First-principles calculation for EIC physics

Our project has utilized the state-of-the-art exascale HPC facilities across the country, including Aurora, Frontier, and Perlmutter.



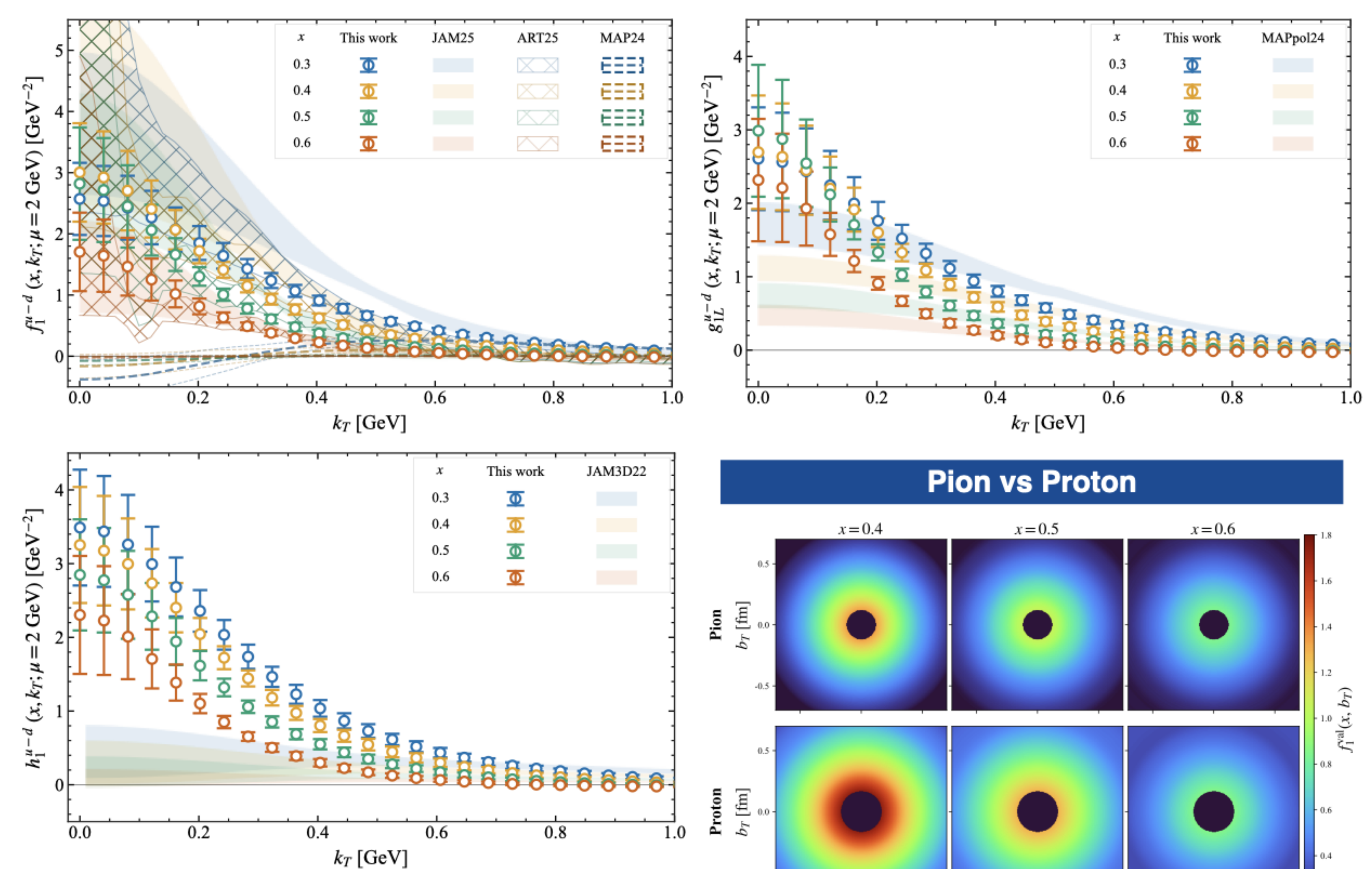
Large Momentum Effective Theory (LaMET) enables precision-controlled determinations of x -dependent parton distributions from lattice QCD.



Proton TMDs at the Physical Point

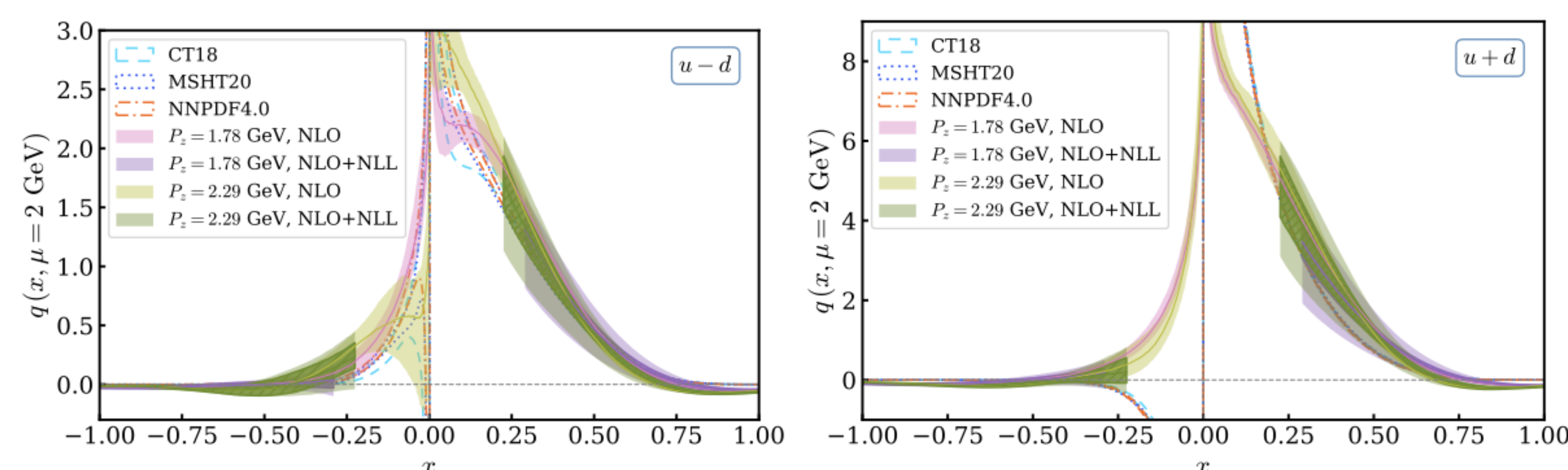
- We simultaneously obtain the unpolarized, helicity and transversity TMDs.
- Tensions among global analyses highlight the importance of **first-principles** lattice calculations as an **independent** determination of TMDPDFs.

First Simultaneous Lattice Calculation of Proton TMDs in Three Polarizations



Proton Unpolarized PDF at the Physical Point

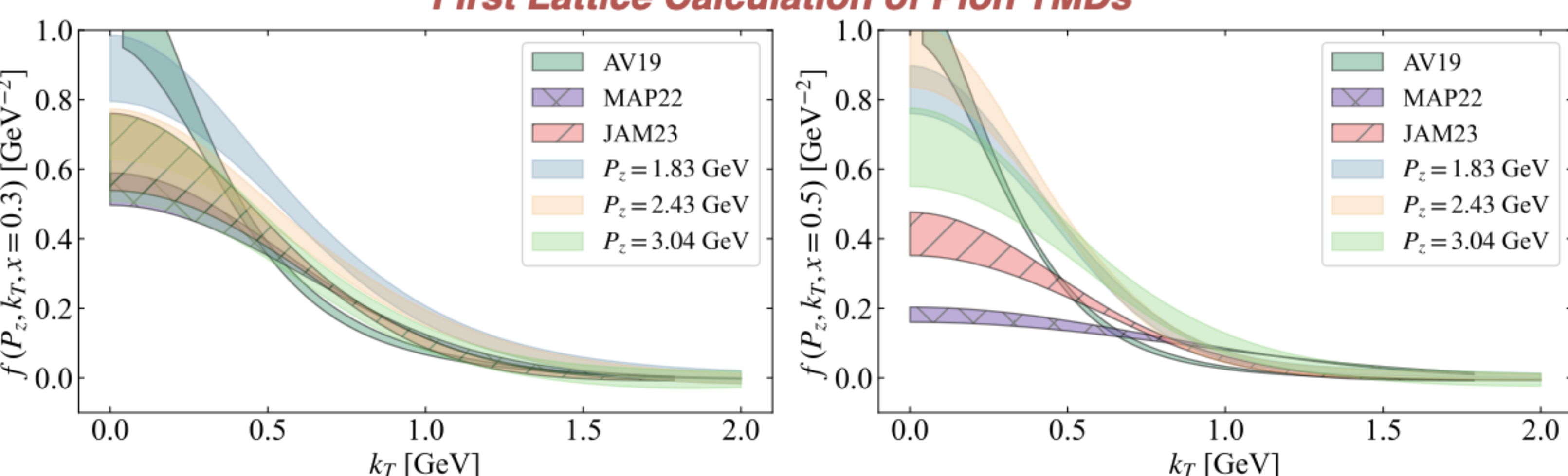
- Physical-point** lattice QCD calculations up to $P_z = 2.29$ GeV yield **convergent** proton PDFs that agree with global analyses in the **moderate- x** region.



Pion Unpolarized TMDs on a Fine Lattice

- Lattice provides predictions at relatively large x , where phenomenological results have large discrepancies with each other.

First Lattice Calculation of Pion TMDs



Other Research Highlights

- Quark Collins-Soper kernel;
- Pion generalized parton distributions (GPDs);
- Pion & Kaon PDFs and light-cone distribution amplitudes (LCDAs);
- Pion & Kaon electric-magnetic form factors at large momentum transfer;
- Hadrons and light nuclei masses and trace anomaly;

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